MultiEMiT: Multimodal Emotion, Irony, and Semiotic Relation Detection in Italian Social TV via Correlation Prompting

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***Abstract*—During live television broadcasts, viewers react on social media with a blend of text, screengrabs, and memes. Decoding affective intent in this stream is hard: posts are short, labels are heavily skewed, and images often twist or invert the written message. We present MultiEMiT, a multimodal dataset of Italian Twitter commentary annotated across five pragmatic dimensions: fine-grained emotions under Plutchik's taxonomy, irony and sarcasm, hate speech, target orientation (broadcast content versus program container), and semiotic text–image relationships (additive, parallel, divergent). We evaluate task-specialized Dual-Encoders (Italian BERT and Vision Transformer with Classifier Chains) against an open-weights generative model, Pixtral 12B. To exploit label dependencies, we introduce correlation prompting: directly injecting empirical cross-task priors into the input text. Visual context improves multi-label emotion recognition, lifting Macro F_1 by $+13\%$ over text-only models. In irony detection, raw visual features introduce semantic noise and degrade performance through the modality gap; injecting correlation priors resolves this misalignment and pushes Dual-Encoder accuracy to $F_1 = 0.740$. Across all tasks, lightweight discriminative models outperform the fine-tuned 12-billion-parameter foundation model. Pixtral suffers from exposure bias and repetitive token loops on long-tail categories, requiring post-hoc truncation ($n = 2$) to eliminate hallucinations.**

***Index Terms*—Affective computing, emotion detection, irony detection, multimodal fusion, Social TV, correlation prompting, vision-language models, Italian NLP.**

I. Introduction

TELEVISION audiences no longer watch broadcasts in silence. During live debates, political talk shows, and reality competitions, viewers actively turn to second-screen platforms like Twitter to share commentary, memes, and screenshots [2]. This stream of real-time viewer feedback offers broadcasters far richer signals than traditional audience ratings (such as Auditel), providing immediate insight into public controversy, approval, and emotional engagement [10].

Capturing these affective signals computationally is difficult. Social TV commentary is short, colloquial, and inherently multimodal. Viewers attach screengrabs and reaction images that do not merely mirror the written text; they exaggerate emotional tone, introduce sarcasm, or invert the literal message completely.

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While multilingual initiatives have advanced text-based emotion analysis, computational resources for Italian remain scarce. Existing Italian benchmarks either focus exclusively on unimodal text [3], [9] or examine general web comments rather than televised social discourse [8]. At the same time, pairing text and images in satirical or reality programming introduces heavy class imbalance and frequent semantic dissonance, confounding standard vision-language models.

To study these dynamics, we introduce **MultiEMiT**, a multimodal dataset of Italian Twitter commentary gathered from television broadcasts. MultiEMiT provides annotations across five pragmatic layers: fine-grained emotions under Plutchik's taxonomy [4], irony and sarcasm, hate speech, target orientation (program container versus broadcast content), and semiotic text–image relationships [5]. Our benchmarking focuses on three primary classification tasks: Emotion Detection, Irony Detection, and Semiotic Relation Detection. The remaining annotations (hate speech and target) serve as contextual signals. To expand training coverage for baseline text encoders, we also draw on the larger unimodal EMiT corpus [3].

Our exploratory analysis uncovered strong empirical correlations between annotation layers: irony correlates with joy, sarcasm pairs with disgust and hostility, and trust associates with program loyalty. Rather than leaving neural models to discover these sparse associations on their own, we propose **Correlation Prompting**. This approach converts empirical Pearson correlation statistics directly into textual prompt prefixes, providing explicit inductive priors during training and inference.

To evaluate these methods, we frame our study around three core research questions:

- **RQ1 (Visual Context):** Does adding visual context help models detect emotion and irony, or do images introduce distracting noise in non-literal posts?
- **RQ2 (Correlation Prompting):** Does injecting statistical label correlations directly into text inputs improve accuracy across discriminative and generative architectures?
- **RQ3 (Architectural Paradigms):** How do compact, task-specialized discriminative models (BERT–ViT Dual-Encoders with Classifier Chains) compare with a 12-billion-parameter foundation model (Pixtral 12B)?

The rest of this paper explores these questions systematically. Section II situates MultiEMiT within Italian affective computing, Social TV analytics, and multimodal document intent. Section III describes the corpus collection, annotation

protocol, and cross-dimensional correlations. Section IV defines our classification tasks, Dual-Encoder architecture, correlation prompting strategy, and Pixtral configurations. Section V presents comparative evaluations and subset analyses across semiotic relations. Section VI summarizes our findings and outlines next steps for real-time broadcast monitoring.

II. Related Works

Most annotated resources for affective computing focus on English. While multilingual initiatives like BRIGHTER [18] aggregate emotion data across 28 languages and 100,000 instances, they omit Italian, pointing users toward standalone external sets like FEEL-IT [9]. This gap leaves Italian social media underrepresented in multi-label benchmarks, despite distinct cultural and linguistic norms in regional political discourse.

Early Italian resources focused strictly on binary or three-class polarity. The first dedicated shift toward fine-grained categories came with *FEEL-IT* [9], which collected 2,037 tweets across politics, sports, and healthcare during 2020. Annotated with Ekman's four basic emotions (anger, joy, sadness, fear), FEEL-IT established that Italian language models like UmBERTo [12] produce reliable affective classifiers. Yet a four-class schema misses mixed feelings and conversational nuances common in viewer reactions. To address broader emotional ranges, *MultiEmotions-It* [8] annotated YouTube and Facebook comments across Plutchik's eight primary emotions [4], polarity, and sarcasm. Their findings showed that while polarity models perform consistently, multi-label emotion recognition and sarcasm detection suffer from severe class sparsity and label ambiguity.

Second-screen platforms have changed how broadcasters analyze viewer engagement [2]. Traditional television ratings (like Auditel) measure raw viewer counts; they cannot gauge whether an audience is laughing, outraged, or sympathetic. The *M.EMO.RAI* project was formed to address this limitation, establishing a joint partnership between the University of Turin and Rai (Radiotelevisione Italiana) [10]. The goal was an automated system to track audience reactions during live broadcasts by ingesting public social network commentary.

This partnership led to the *EMiT* (Emotions in Italian) corpus, benchmarked as an EVALITA 2023 shared task [3]. With 5,966 social media posts from live broadcasts (including the Sanremo Festival and scripted drama), EMiT adopted Plutchik's taxonomy. It also introduced the *Target* dimension, distinguishing whether a comment attacks the broadcast *Content* (the topic, guest, or political issue) or evaluates the *Container* (the host, studio format, or network production). EVALITA benchmark results showed that while classifiers perform well on familiar shows, accuracy drops on out-of-domain broadcasts. We use EMiT as an external textual corpus to pre-train and enrich our baseline emotion encoders.

Irony detection requires models to recognize when a speaker's intended meaning opposes their literal words. In Italian NLP, the standard benchmark is *IronITA* [11], curated for EVALITA 2018 with nearly 5,000 tweets annotated for irony and sarcasm (defined as hostile or denigratory mockery). Results on IronITA showed that text-only classifiers struggle on short posts when pragmatic context is missing. To provide our models with domain-adapted pragmatic competence, we use IronITA to fine-tune UmBERTo before evaluating on television commentary.

Linguistic text only tells half the story in modern social media. In their study on document intent, Kruk et al. [5] demonstrated that images in social posts rarely just illustrate captions; they alter communicative intent through distinct semiotic relationships, ranging from complementary addition to direct divergence. Incongruent or divergent pairings generate satirical humor and memes, where the image subverts the text. Kruk et al. studied English Instagram posts. The semiotic relationship between text and visual media in Italian political satire and reality television remains largely unstudied. MultiEMiT applies this semiotic framework directly to Italian Social TV.

III. Dataset Analysis

We introduce the MultiEMiT corpus, a multi-layered multimodal dataset of Italian Twitter commentary gathered during live television broadcasts. The data originates from the M.EMO.RAI project [10], which set out to track audience emotional responses in real time. Rather than restricting annotations to positive or negative sentiment, MultiEMiT applies an expanded eight-emotion taxonomy derived from Plutchik's psychoevolutionary model [4].

MultiEMiT draws from two television broadcasts representing contrasting formats and registers:

ProSAT: Twitter commentary on *Propaganda Live*, a satirical talk show on LA7 combining investigative journalism and political commentary [6]. The collection captures the broadcast of November 19, 2021, when host Diego Bianchi reported live from the Poland–Belarus border during a humanitarian crisis. The gravity of the event prompted solemn reactions dominated by indignation, grief, and civic debate.

EMiT (Pechino Express): Collected as an extension of the EVALITA EMiT benchmark [3], this sub-corpus covers *Pechino Express*, an adventure reality show broadcast on Rai2 [7]. Collected during the season premiere on February 11, 2020, viewer discourse centers on lighthearted entertainment, contestant rivalries, and humorous commentary.

Both partitions share an identical structure: a purely textual partition and a multimodal partition pairing the text with its attached media (screenshots, edited images, and memes). This paired structure allows direct comparison between unimodal and multimodal models, showing where visual content modulates the written post. In addition to MultiEMiT, we use the external textual EMiT corpus (5,966 posts across multiple programs [3]) as supplementary training data for baseline emotion encoders.

Annotation was conducted on Sensitag, a web annotation platform developed at the University of Turin. Each post was evaluated independently by three annotators with backgrounds in computer science or linguistics. Final labels were assigned by majority vote (at least two out of three annotators agreeing).

The corpus is annotated across five pragmatic dimensions:

- **Emotions:** Ten affective categories based on Plutchik's model [4] (*Joy–Sadness*, *Trust–Disgust*, *Fear–Anger*, *Surprise–Anticipation*), supplemented with *Love* and *Neutral*. This task is multi-label, allowing posts to carry multiple concurrent emotional states.
- **Irony and Sarcasm:** Three exclusive classes separating literal statements (*Unironic*), humorous figurative inversion (*Irony*), and hostile, denigratory mockery (*Sarcasm*).
- **Hate Speech:** Identifies abusive language, insults, or harassment directed at individuals or protected groups.
- **Target:** Distinguishes whether comments address the broadcast *Content* (political issues, guests, or news events) or the *Container* (the studio host, network, or production execution).
- **Semiotic Relation:** Categorizes multimodal document intent [5] into *Additive* (modality enrichment), *Parallel* (redundant information), and *Divergent* (ironic contrast or meme dissonance).

Sarcasm and emotion in short social media posts are inherently ambiguous. Read in isolation, a comment on the border crisis such as *"Terrible, I am ashamed to be European"* could be seen as shame (*Disgust* + *Fear*), contempt (*Anger* + *Disgust*), or disapproval (*Surprise* + *Sadness*).

Providing annotators with the attached image resolved much of this ambiguity, substantially lifting inter-annotator agreement (Fleiss' κ) across most emotion labels. Seeing photos of shivering refugees grounded the context, directing consensus toward sadness and distress. This empirical finding provides direct justification for multimodal modeling: if visual context clarifies ambiguous intent for human annotators, multimodal neural architectures should benefit from the same disambiguating cues.

Following gold-standard consolidation, the MultiEMiT corpus resulted in two main partitions: a purely **Textual** dataset comprising 2,909 posts, and a **Multimodal** dataset comprising 645 posts with paired image attachments.

As typical of social media discourse, label distributions exhibit pronounced class imbalance. Tables I and II present the complete distribution of labels across both partitions, broken down by broadcast sub-corpus.

Across both partitions, the most frequent affective states are *Trust* and *Joy*, indicating an overall positive engagement with broadcast programming. However, negative emotions exhibit marked domain dependency: *Sadness*, *Anger*, and *Fear* are heavily concentrated within ProSAT due to the geopolitical reportage, whereas they remain negligible in the lighthearted Pechino Express subset.

Similarly, irony displays genre-specific dynamics. In the multimodal corpus, irony is prevalent in Pechino Express (88 instances) despite Propaganda Live being a satirical format. This occurs because the specific Propaganda Live episode under study addressed an acute humanitarian crisis, prompting solemn reflection rather than playful sarcasm. Regarding semiotic relations, the *Additive* relation dominates in both subsets, while *Parallel* and *Divergent* relations appear far more frequently in entertainment contexts where visual memes thrive.

TABLE I
LABEL DISTRIBUTION IN THE MULTIEMIT TEXTUAL DATASET (2,909 POSTS)

Label	Total (2,909)	ProSAT	EMiT
Emotions (multi-label)			
Love	340	119	221
Anticipation	201	75	126
Disgust	410	207	203
Trust	613	281	332
Joy	552	257	295
Neutral	321	31	290
Fear	55	44	11
Anger	163	121	42
Surprise	486	225	261
Sadness	255	211	44
Irony			
Irony	481	237	244
Unironic	2,072	828	1,244
Sarcasm	41	15	26
Offensiveness			
Offensive	52	31	21
Target			
Container	1,147	343	804
Content	1,138	710	428

TABLE II
LABEL DISTRIBUTION IN THE MULTIEMIT MULTIMODAL DATASET (645 POSTS)

Label	Total (645)	ProSAT	EMiT
Emotions (multi-label)			
Love	50	17	33
Anticipation	61	10	51
Disgust	78	18	60
Trust	214	82	132
Joy	168	79	89
Neutral	90	21	69
Fear	4	4	0
Anger	34	29	5
Surprise	82	24	58
Sadness	25	23	2
Irony			
Irony	109	21	88
Unironic	516	199	317
Sarcasm	20	5	15
Offensiveness			
Offensive	6	3	3
Target			
Container	395	103	292
Content	199	85	114
Semiotic Relation			
Additive	389	158	231
Divergent	158	42	116
Parallel	100	10	90

The multi-layered annotation of MultiEMiT enables systematic investigation into the statistical dependencies connecting emotional, figurative, and semiotic dimensions. By computing Pearson correlation coefficients (r) across all label pairs, we identified pronounced interdependencies indicating that pragmatic categories act as strong mutual predictors.

Examining the correlation matrix for **Emotions** (Fig. 3), a clear dichotomy emerges between the *Container* and *Content* targets. Program-directed affective states (*Trust* $r = 0.229$, *Anticipation* $r = 0.181$) correlate positively with Container, indicating audience loyalty toward the show. Conversely, content-reactive emotions (*Surprise* $r = 0.319$, *Sadness* $r = 0.214$, *Disgust* $r = 0.205$) associate with Content. Sarcasm also

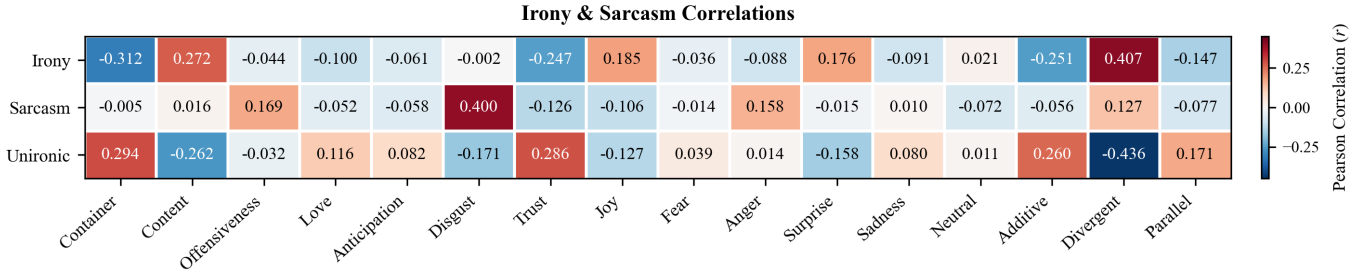


Fig. 1. Pearson correlation matrix for the irony and sarcasm dimension across multi-level annotations.

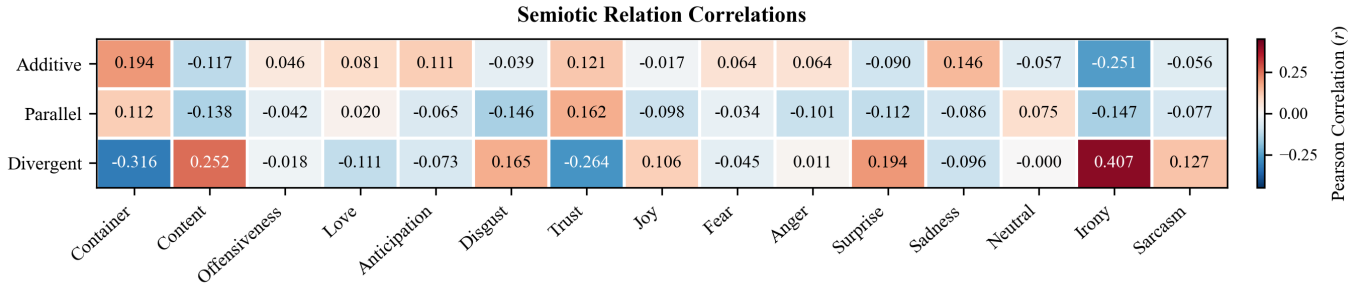


Fig. 2. Pearson correlation matrix for semiotic text–image relations against contextual and affective markers.

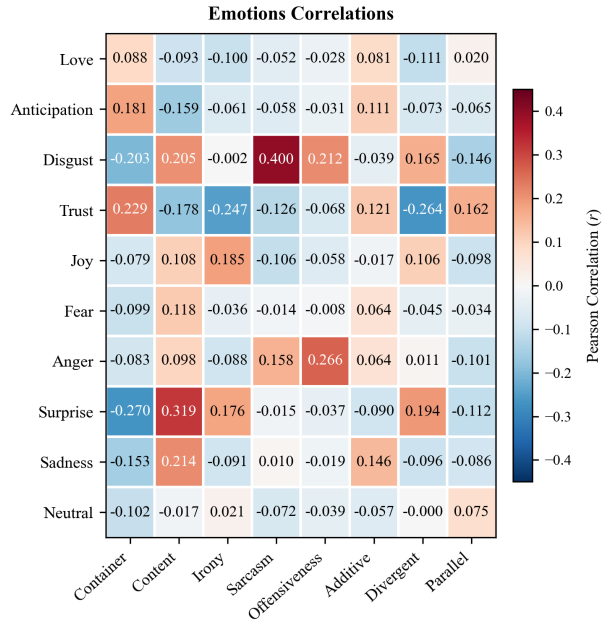


Fig. 3. Emotions Correlations: Pearson correlation matrix between emotion labels and contextual dimensions (broadcast targets, figurative speech, and semiotic relations).

reveals a hostile pragmatic pattern, correlating positively with *Disgust* ($r = 0.400$), *Anger* ($r = 0.158$), and offensive language ($r = 0.212$).

Analyzing **Irony** (Fig. 1), we observe that figurative speech relies fundamentally on semiotic contrast. Irony correlates strongly with *Divergent* text–image relations ($r = 0.407$) and the *Content* target ($r = 0.272$), demonstrating that multimodal irony frequently arises from intentional tension between the visual image and the written caption. Sincere posts (*Unironic*), by contrast, correlate with *Container* ($r = 0.294$), *Trust* ($r = 0.286$), and complementary visual relations (*Additive* $r = 0.260$, *Parallel* $r = 0.171$).

Semiotic relations (Fig. 2) confirm that divergent pairings drive social media irony: Divergent posts correlate positively with *Irony* ($r = 0.407$), *Content* ($r = 0.252$), and *Surprise* ($r = 0.194$). In contrast, the *Additive* relation associates with *Container* ($r = 0.194$) and narrative elaboration, while the *Parallel* relation exhibits minimal correlation with complex emotional states, reflecting literal and redundant text–image usage.

IV. Experimental Design

This section details the experimental methodology established to evaluate affective classification on the MultiEMiT corpus. We evaluate three distinct tasks: Emotion Detection, Irony Detection, and Semiotic Relation Detection. The experiments measure the performance differences between unimodal textual baselines, a discriminative multimodal architecture (Dual-Encoder), and a generative foundation model (Pixtral 12B), while assessing the empirical impact of correlation prompting across architectures.

We formalize three classification benchmarks corresponding to different levels of communicative intent:

- **Emotion Detection:** Multi-label classification over ten candidate emotions (*Anger*, *Anticipation*, *Disgust*, *Fear*, *Joy*, *Love*, *Neutral*, *Sadness*, *Surprise*, and *Trust*). Models predict any valid subset of concurrent emotions using binary cross-entropy or binary chains.

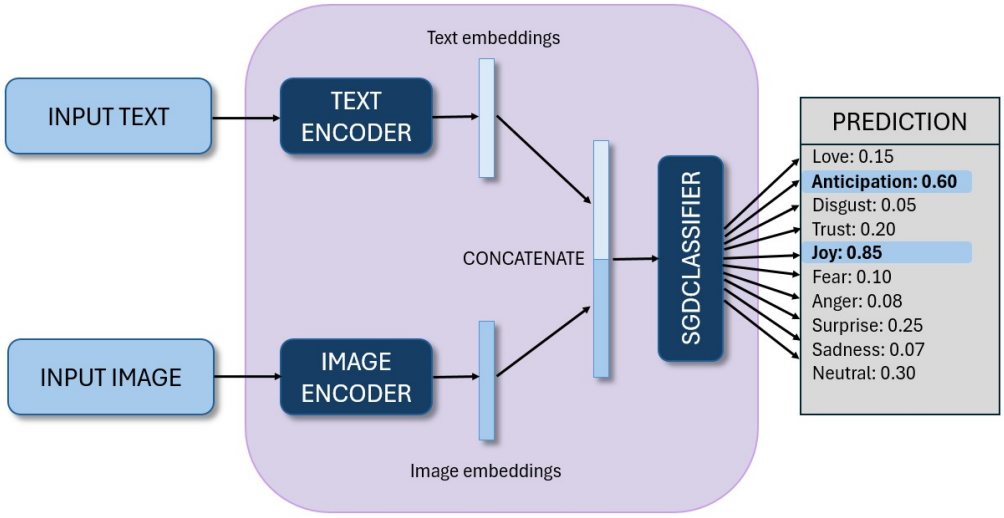


Fig. 4. Dual-Encoder multimodal architecture. Text and image representations are extracted in parallel via domain-adapted BERT and ViT encoders, concatenated into a joint vector ($d = 1536$), and passed to a Classifier Chain head.

- **Irony Detection:** Single-label classification categorizing each post into one of three mutually exclusive classes: *Unironic*, *Irony*, or *Sarcasm*.
- **Semiotic Relation Detection:** Single-label classification identifying the semantic text–image interaction as *Additive*, *Parallel*, or *Divergent*.

Due to heavy class skew, raw accuracy paints a misleading picture of model quality. We therefore evaluate systems using **Precision**, **Recall**, and the **F1-score**. In overall comparisons across all three tasks, the **Macro F1-score** serves as the primary metric, weighting each class equally to penalize poor performance on minority labels. In fine-grained subset evaluations across semiotic relations (Section V), we report the **Micro F1-score** to maintain metric stability when sample sizes in individual subsets are small.

To establish strong text-only reference points, we evaluated models built on the RoBERTa architecture. For Irony and Semiotic Relation detection, we used *UmBERTo* [12], a RoBERTa-based language model pre-trained on Italian Common Crawl with whole-word masking. For Irony detection, UmBERTo was first fine-tuned on the IronITA benchmark [11] before adaptation to MultiEMiT. For Emotion Detection, we used *FEEL-IT* [9], fine-tuned on the combined MultiEMiT textual split and the external EMiT dataset [3] to adapt its representation space to our ten-class taxonomy. These text models process the tweet text alone without visual inputs.

To capture cross-modal interactions, we implemented a discriminative Dual-Encoder pipeline (Fig. 4). The textual modality is processed by the domain-adapted language encoder (FEEL-IT for emotions, UmBERTo for irony and relations), while the visual modality is processed in parallel by a Vision Transformer (ViT-base-patch16-224) [14]. Both unimodal encoders extract high-level representations: the 768-dimensional '[CLS]' token embedding from the text transformer and the 768-dimensional pooled representation from the ViT are concatenated into a 1536-dimensional joint multimodal vector.

To predict class labels from this joint representation, we employ Classifier Chains [16]. This architecture models label interdependencies by sequentially conditioning subsequent binary classifiers on previous predictions, making it well suited for multi-label emotion spaces. Specifically, we configure Scikit-Learn's `ClassifierChain` wrapping an `SGDClassifier` with logistic loss, L_2 regularization, and fixed random seeding. This linear head projects the concatenated latent space to class predictions without overfitting on the compact multimodal split.

Motivated by the strong label dependencies discovered during corpus analysis (Section III), we developed a structured *Correlation Prompting* strategy. Rather than hoping the neural network will infer subtle cross-task correlations from scarce training examples, we inject empirical Pearson correlation coefficients directly into the input text as natural language context. For example, in the Irony detection task, the input tweet is appended with: *"Statistics: Sarcasm correlates positively with Disgust and Anger. Irony correlates positively with Divergent relation."*

To evaluate model sensitivity to the volume of injected priors, we structured prompts across three distinct correlation tiers based on Pearson coefficient thresholds: (i) **Short prompts**, incorporating only strong associations with $|r| \geq 0.25$; (ii) **Medium prompts**, including moderate correlations with $|r| \geq 0.20$; and (iii) **Long prompts**, providing detailed contextual priors with all correlations satisfying $|r| \geq 0.15$. All prompt instructions were formulated in English, following findings from the BRIGHTER benchmark [18] indicating that multilingual LLMs adhere better to instructions and exhibit lower variance when prompted in English rather than the target language.

To compare specialized discriminative systems against mod-

ern generative vision-language models, we benchmarked **Pixtral 12B** [17], an open-weights multimodal language model based on Mistral Nemo featuring a dedicated 400M-parameter vision encoder capable of handling arbitrary image resolutions. We evaluated Pixtral across three experimental regimes:

- **Zero-Shot Prompting:** Out-of-the-box reasoning across prompt tiers (Short, Medium, and three Long prompt formulations: *long_v1*, *long_v2*, *long_v3*), testing generation with and without correlation context.
- **Few-Shot (Three-Shot) Prompting:** In-context learning for Semiotic Relation Detection, providing three complete multimodal exemplars (one per relation category) within the prompt window.
- **LoRA Fine-Tuning:** Low-Rank Adaptation on the MultiEMiT multimodal training set ($r = 16$, $\alpha = 32$, RSLoRA, all linear target modules, dropout 0.1). This configuration adapts 66,322,432 parameters out of 12.75 billion (0.52% trainable parameters). Training ran with per-device batch size 1, 8 gradient accumulation steps (effective batch size 8), learning rate 3×10^{-5}, FP16 mixed precision, and gradient checkpointing over 2 epochs.

All generative experiments were evaluated over three independent runs at two decoding temperatures: $T = 0.1$ (greedy, low-variance decoding) and $T = 0.9$ (stochastic exploratory decoding).

Unlike discriminative encoders that output bounded probability vectors over predefined classes, Pixtral generates labels autoregressively. During multi-label emotion fine-tuning, ranking bias and exposure bias degraded the stopping token (<EOS>), causing repetitive token loops and out-of-vocabulary synonyms (e.g., generating *Joy, Joy, Happiness* instead of *Joy*). To resolve this without retraining, we implemented a post-processing truncation parameter n, restricting extraction to the first n unique valid taxonomy emotions generated. Setting $n = 2$ matches the empirical ground truth and eliminates unconstrained tail hallucinations.

V. Results and Discussion

This section presents the empirical evaluation of the unimodal and multimodal architectures across the three tasks: Emotion Detection, Irony Detection, and Semiotic Relation Detection. We analyze the quantitative impact of integrating visual features (comparing blind text baselines to the Dual-Encoder), the performance gains induced by correlation prompting, and the strengths and failure modes of Pixtral 12B. We also examine model behavior disaggregated across semiotic relational subsets.

Emotion detection represents the most challenging benchmark due to the high cardinality of the ten-class label space and the multi-label nature of affective expression in social media.

On the textual dataset, the baseline discriminative model (FEEL-IT fine-tuned on MultiEMiT and EMiT) achieved a Macro F_1-score of 0.460, increasing to 0.540 when trained with correlation prompting, and reaching 0.590 when retrained across two isolated phases with correlation context. When evaluating exclusively on the multimodal test set, the blind text-only baseline (evaluating only the text of multimodal posts) achieves a Macro F_1-score of 0.300 without correlations and 0.370 with correlations.

TABLE III
Complete Comparison: Pixtral 12B versus Dual-Encoder in Emotion Detection ($T = 0.1$ and $T = 0.9$)

Model	Configuration	T = 0.1 F1	T = 0.9 F1
Discriminative Multimodal (Dual-Encoder)			
BERT-ViT	FT (With Corr.)	**0.450**	**0.450**
BERT-ViT	FT (No Corr.)	0.430	0.430
Generative (Pixtral 12B)			
Pixtral (long_v1)	FT ($n = 2$)	**0.426**	0.359
Pixtral (short)	FT ($n = 2$)	0.397	**0.373**
Pixtral (short)	FT (all predictions)	0.318	0.312
Pixtral (short)	ZS (With Corr.)	0.291	0.251
Pixtral (long_v1)	ZS (With Corr.)	0.275	0.266
Pixtral (medium)	ZS (With Corr.)	0.269	0.262
Pixtral (long_v2)	ZS (No Corr.)	0.266	0.237
Pixtral (long_v1)	ZS (No Corr.)	0.261	0.261
Pixtral (short)	ZS (No Corr.)	0.261	0.291
Pixtral (long_v3)	ZS (With Corr.)	0.258	0.262
Pixtral (medium)	ZS (No Corr.)	0.254	0.250
Pixtral (long_v2)	ZS (With Corr.)	0.253	0.286
Pixtral (long_v1)	FT (all predictions)	0.238	0.290
Pixtral (long_v3)	ZS (No Corr.)	0.220	0.215

TABLE IV
Impact of Truncation Parameter (n) on Pixtral Emotion Detection Performance

Number of Predictions (n)	Mean Macro F1
1	0.338
2	**0.389**
3	0.339
4	0.325
5	0.308

Introducing visual features via the Dual-Encoder architecture substantially enhances classification performance: the multimodal system achieves a Macro F_1-score of 0.430 without correlations and 0.450 with correlations. This marked improvement (+13% absolute over the blind unprompted text baseline) empirically confirms that visual context provides essential disambiguating cues for recognizing emotional expressions in social commentary, with correlation prompting providing a complementary gain.

Turning to generative foundation models, fine-tuned Pixtral 12B achieved an F_1-score of 0.426 ($T = 0.1$). While competitive with the discriminative Dual-Encoder without correlations (0.430), it remains below the correlation-prompted Dual-Encoder (0.450), showing that task-adapted discriminative representations remain superior for multi-label Italian emotion classification. Complete comparisons across prompt tiers and temperatures are reported in Table III.

During Pixtral fine-tuning, autoregressive decoding generated repetitive loops and extra-taxonomical tokens when predicting emotion lists. However, qualitative analysis revealed that predicted emotions were ordered by descending model confidence. Implementing the truncation parameter n to retain the top-n unique valid emotions effectively eliminated generative noise. As detailed in Table IV, performance peaks distinctly at $n = 2$ (Mean Macro $F_1 = 0.389$). Allowing unconstrained generation ($n \geq 5$) degrades performance to 0.308, as spurious tail predictions incur heavy false-positive

TABLE V
EMOTION DETECTION MICRO F1-SCORE ACROSS SEMIOTIC SUBSETS

Model Setting	ADD	PAR	DIV
Dual-Encoder (BERT–ViT)			
No correlations	0.47	0.52	0.38
With correlations	**0.54**	**0.60**	0.42
Generative LLM (Pixtral 12B)			
Zero-Shot (No corr.)	0.29	0.23	**0.44**
Zero-Shot (With corr.)	0.32	0.30	0.43
Fine-tuned (No corr.)	0.33	0.41	0.41

TABLE VI
COMPLETE COMPARISON: PIXTRAL 12B VERSUS DUAL-ENCODER IN IRONY DETECTION ($T = 0.1$ AND $T = 0.9$)

Model	Configuration	T = 0.1 F1	T = 0.9 F1
Discriminative Multimodal (Dual-Encoder)			
BERT-ViT	FT (With Corr.)	**0.740**	**0.740**
BERT-ViT	FT (No Corr.)	0.630	0.630
Generative (Pixtral 12B)			
Pixtral (long_v2)	ZS (With Corr.)	**0.512**	0.396
Pixtral (long_v2)	FT (all predictions)	0.405	**0.454**
Pixtral (short)	ZS (With Corr.)	0.348	0.298
Pixtral (long_v3)	ZS (With Corr.)	0.340	0.281
Pixtral (short)	FT (all predictions)	0.329	0.394
Pixtral (medium)	ZS (With Corr.)	0.264	0.270
Pixtral (long_v1)	ZS (With Corr.)	0.248	0.295
Pixtral (long_v2)	ZS (No Corr.)	0.165	0.218
Pixtral (long_v3)	ZS (No Corr.)	0.148	0.198
Pixtral (long_v1)	ZS (No Corr.)	0.126	0.158
Pixtral (short)	ZS (No Corr.)	0.054	0.108
Pixtral (medium)	ZS (No Corr.)	0.051	0.137

penalties. Setting $n = 2$ matches the empirical ground truth, where the vast majority of posts express one or two concurrent emotions.

Disaggregating model performance across the Additive (ADD), Parallel (PAR), and Divergent (DIV) semiotic partitions reveals striking structural variations (Table V).

Both model families experience their lowest scores on the *Divergent* subset (Micro $F_1 = 0.38$ to 0.42). In divergent posts, intentional tension between text and imagery disrupts emotional coherence, confounding automated classifiers. Conversely, models perform best on the *Parallel* partition ($F_1 = 0.60$ for the prompted Dual-Encoder), where redundant visual information reinforces literal textual meaning. For the dominant *Additive* subset, correlation prompting delivers a $+0.07$ boost for the Dual-Encoder, demonstrating that statistical priors help integrate complementary cross-modal evidence.

Irony detection presents an inverted dynamic compared to emotion recognition. In figurative discourse, visual content frequently introduces pragmatic incongruity rather than semantic convergence. Table VI summarizes comparative results.

When evaluated on the multimodal test set, the blind textual baseline achieves an F_1-score of 0.670 without correlations and 0.710 with correlations. In sharp contrast, the multimodal Dual-Encoder without correlations achieves an F_1-score of only 0.630, underperforming the blind text model by -0.040. This degradation points to a core multimodal challenge: unguided visual features act as semantic noise. Because irony relies on incongruity, raw visual embeddings pull the multimodal vector away from the textual signal. This pattern aligns with findings on the *Modality Gap* [19]: concatenating representations from separately pre-trained encoders (UmBERTo and ViT) without joint cross-modal pre-training leaves a geometric misalignment in latent space.

Injecting statistical correlations reverses this drop: the correlation-prompted Dual-Encoder reaches an F_1-score of **0.740**, outperforming both the blind textual baseline (0.710) and unprompted multimodal fusion (0.630). Explicit priors help the classifier treat visual dissonance as evidence of irony rather than noise.

In the generative regime, Pixtral 12B struggles severely in zero-shot settings without correlations (F_1 ranging between 0.051 and 0.165 across prompt variants at $T = 0.1$). However, incorporating correlation prompts triggers a dramatic recovery, elevating zero-shot performance to 0.512 with the *long_v2* prompt (+0.347 gain). After LoRA fine-tuning, Pixtral achieves an F_1-score of 0.454 ($T = 0.9$). Although competitive, it remains substantially below the specialized Dual-Encoder (0.740).

Examining irony detection across semiotic subsets (Table VII) elucidates how models process multimodal dissonance.

TABLE VII
IRONY DETECTION MICRO F1-SCORE ACROSS SEMIOTIC SUBSETS

Model Setting	ADD	PAR	DIV
Dual-Encoder (BERT–ViT)			
No correlations	0.83	**1.00**	0.55
With correlations	**0.88**	**1.00**	**0.68**
Generative LLM (Pixtral 12B)			
Zero-Shot (No corr.)	0.14	0.00	0.18
Zero-Shot (With corr.)	0.57	0.78	0.57
Fine-tuned (No corr.)	0.87	**1.00**	0.58

On the *Parallel* partition, both the Dual-Encoder and fine-tuned Pixtral achieve a perfect score of 1.00. This outcome arises because the gold multimodal test set contains zero ironic posts with a parallel relation; the trained models correctly classify all instances as sincere (*Unironic*). Conversely, unprompted zero-shot Pixtral collapses on parallel instances (0.00), hallucinating false-positive irony. On the challenging *Divergent* subset, where irony is concentrated, the Dual-Encoder without correlations attains only 0.55; injecting correlation prompting raises this to 0.68 (+0.13 gain), confirming that correlation priors are vital for decoding divergent visual irony.

Semiotic relation detection requires the model to directly categorize the semantic interaction between modalities into Additive, Parallel, or Divergent. Table VIII presents the complete evaluation.

The Dual-Encoder with correlation prompting achieves the highest performance across all configurations (Macro $F_1 = 0.690$), outperforming its unprompted counterpart (0.640). Fine-tuned Pixtral peaks at 0.613 ($T = 0.1$), while zero-shot prompted variants reach 0.517. Although Pixtral possesses vast pre-trained visual knowledge, discerning abstract semiotic relationships remains challenging for generalist vision-language models, whereas task-specific discriminative classification heads with Classifier Chains effectively map the cross-modal feature space.

TABLE VIII
COMPLETE COMPARISON: PIXTRAL 12B VERSUS DUAL-ENCODER IN SEMIOTIC RELATION DETECTION ($T = 0.1$ AND $T = 0.9$)

Model	Configuration	T = 0.1 F1	T = 0.9 F1
Discriminative Multimodal (Dual-Encoder)			
BERT-ViT	FT (With Corr.)	**0.690**	**0.690**
BERT-ViT	FT (No Corr.)	0.640	0.640
Generative (Pixtral 12B)			
Pixtral (long_v1)	FT (all predictions)	**0.613**	**0.574**
Pixtral (short)	FT (all predictions)	0.555	0.546
Pixtral (long_v3)	ZS (With Corr.)	0.517	0.486
Pixtral (long_v3)	TS (With Corr.)	0.487	0.508
Pixtral (long_v2)	TS (With Corr.)	0.483	0.530
Pixtral (medium)	TS (With Corr.)	0.480	0.469
Pixtral (long_v1)	TS (With Corr.)	0.467	0.495
Pixtral (medium)	ZS (With Corr.)	0.429	0.414
Pixtral (long_v2)	TS (No Corr.)	0.424	0.402
Pixtral (medium)	TS (No Corr.)	0.403	0.381
Pixtral (long_v1)	ZS (With Corr.)	0.394	0.409
Pixtral (medium)	ZS (No Corr.)	0.388	0.359
Pixtral (long_v3)	TS (No Corr.)	0.387	0.359
Pixtral (long_v1)	ZS (No Corr.)	0.387	0.408
Pixtral (short)	TS (With Corr.)	0.387	0.448
Pixtral (long_v2)	ZS (With Corr.)	0.382	0.367
Pixtral (long_v1)	TS (No Corr.)	0.369	0.414
Pixtral (short)	ZS (With Corr.)	0.359	0.392
Pixtral (short)	TS (No Corr.)	0.355	0.378
Pixtral (long_v3)	ZS (No Corr.)	0.344	0.391
Pixtral (long_v2)	ZS (No Corr.)	0.323	0.315
Pixtral (short)	ZS (No Corr.)	0.242	0.304

TABLE IX
SEMIOTIC RELATION DETECTION MICRO F1-SCORE ACROSS TARGET CLASSES

Model Setting	ADD	PAR	DIV
Custom Baselines (UmBERTo and Dual-Encoder)			
Text-only (No corr.)	0.65	0.53	0.61
Text+Image (No corr.)	0.73	**0.61**	0.59
Text-only (With corr.)	0.75	0.54	0.68
Text+Image (With corr.)	**0.77**	0.59	**0.72**
Generative LLM (Pixtral 12B – long_v1)			
Zero-Shot (No corr.)	0.62	0.00	0.56
Zero-Shot (With corr.)	0.61	0.00	0.56
Three-Shot (With corr.)	0.60	0.17	0.64
Fine-tuned (No corr.)	0.71	0.28	0.60

Disaggregating performance across individual relation categories (Table IX) highlights architectural differences under class imbalance.

The Dual-Encoder exhibits balanced competence across classes, reaching 0.77 on Additive, 0.59 on Parallel, and 0.72 on Divergent. Conversely, Pixtral fails on the minority *Parallel* class (scoring 0.00 in zero-shot and 0.28 under fine-tuning), repeatedly defaulting to the majority class (Additive) due to autoregressive frequency bias. Discriminative classifiers with balanced class weighting maintain stable decision thresholds despite this acute skew.

VI. CONCLUSION

We presented MultiEMiT, a multimodal dataset of Italian Twitter commentary annotated across five pragmatic layers: Plutchik emotions, irony, hate speech, target, and semiotic relations. Benchmarking specialized Dual-Encoders against Pixtral 12B revealed three key findings:

- **RQ1 (Visual Context):** Imagery resolves textual ambiguity when modalities align, lifting multi-label emotion recognition by $+13\%$ Macro F_1 over blind text baselines. In irony, unguided visual features introduce semantic noise, degrading performance through the modality gap.
- **RQ2 (Correlation Prompting):** Injecting empirical correlation priors as textual prefixes consistently improves accuracy. For irony, correlation prompting bridges the cross-modal representation, lifting the Dual-Encoder to peak performance ($F_1 = 0.740$).
- **RQ3 (Architectural Paradigms):** Compact discriminative models (BERT–ViT with Classifier Chains) outperform the 12B foundation model on domain commentary. Pixtral exhibits exposure bias on long-tail classes, requiring post-hoc truncation ($n = 2$) to curb hallucinations.

Looking ahead, our findings open two promising avenues for future research. First, we plan to pursue domain expansion across television genres, extending MultiEMiT beyond political talk shows (*Propaganda Live*) and reality television (*Pechino Express*) to scripted drama, live sports, and cultural broadcasts to overcome genre-specific class skew and evaluate broader cross-genre generalizability. Second, we aim to investigate affective annotation schemas tailored specifically to Social TV dynamics. Rather than relying solely on generalized categorical taxonomies like Plutchik's, future schemas could target engagement-driven emotions (*hype*, *amusement*, *frustration*, *empathy*, *boredom*) alongside viewer stances (*support*, *opposition*, *hatewatching*, *shipping*, *neutral*) to capture the nuanced communicative behaviors of digital television audiences.

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